

Technical note

Electrospinning of chitosan dissolved in concentrated acetic acid solution

Xinying Geng, Oh-Hyeong Kwon, Jinho Jang*

School of Advanced Materials and System Engineering, Kumoh National Institute of Technology, 1 Yongho-dong Kumi, Kyungbuk 730-701, Republic of Korea

Received 10 December 2004; accepted 19 January 2005

Available online 19 March 2005

Abstract

Chitosan nanofibers were electrospun from aqueous chitosan solution using concentrated acetic acid solution as a solvent. A uniform nanofibrous mat of average fiber diameter of 130 nm was obtained from the following optimum condition: 7% chitosan solution in aqueous 90% acetic acid solution was successfully electrospun in the electric field of 4 kV/cm. The aqueous acetic acid concentration higher than 30% was prerequisite for chitosan nanofiber formation, because more concentrated acetic acid in water progressively decreased surface tension of the chitosan solution and concomitantly increased charge density of jet without significant effect on solution viscosity. However, acetic acid solution more than 90% did not dissolve enough chitosan to make spinnable viscous concentration. Only chitosan of a molecular weight of 106,000 g/mol produced bead-free chitosan nanofibers, while low- or high-molecular-weight chitosans of 30,000 and 398,000 g/mol did not. Average fiber diameters and size distribution decreased with increasing electric field and more bead defects appeared at 5 kV/cm or more.

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Keywords: Chitosan; Nanofiber; Electrospinning; Acetic acid; Surface tension; Charge density**1. Introduction**

Electrospinning has been recognized as an efficient method for the fabrication of submicron-sized fibers and various macromolecules have been electrospun into ultrafine fibers as thin as several nanometers [1–4]. When an electric field is applied between a needle capillary end and a collector, surface charge is induced on a polymer fluid deforming a spherical pendant droplet to a conical shape. As the electric field surpasses a threshold value where electrostatic repulsion force of surface charges overcome surface tension, the charged fluid jet is ejected from the tip of the Taylor cone and the charge density on the jet interacts with the external field to produce an instability [3]. High surface charge

densities enhance a whipping mode rather than axisymmetric mode, where bending of the jet produces highly stretched polymeric fiber with simultaneous rapid evaporation of the solvent [4]. Important parameters in electrospinning are not only polymer and solution properties such as molecular weight, viscosity, conductivity and surface tension, but also electrospinning conditions such as applied electric voltage, tip-to-collector distance, feeding rate, etc. [5–7]. Nanofibers have amazing characteristics such as very large surface area-to-volume ratio and high porosity with very small pore size. Therefore, nanofibers can be promising materials for many biomedical applications such as tissue templates, medical prostheses, artificial organ, wound dressing, drug delivery, pharmaceutical composition [5–7]. Several biomacromolecular nanofibers, such as DNA [8], silk [9–11], collagen [12,13], fibrinogen [14], gelatin [15], hyaluronic acid [16] and oxidized cellulose [17], have been electrospun from solvent solutions using

*Corresponding author. Tel.: +82 54 478 7715;
fax: +82 54 478 7710.

E-mail address: jh.jang@kumoh.ac.kr (J. Jang).

trifluoroethanol, hexafluoro-2-propanol, formic acid, etc. It seems more difficult to produce biomacromolecular nanofibers due to their limited solubility to most organic solvents, ionic character in dissolved state and three-dimensional networks of strong hydrogen bonds. Chitosan is an *N*-deacetylated product of chitin, the second-most abundant natural polysaccharide next to cellulose, which is embedded in a protein matrix of a crustacean shell or a squid pen [18]. Chitosan has many useful properties such as biocompatibility, biodegradability, antimicrobial activity, wound healing property, antitumor effect, etc. Electrospinning of chitosan solution has been reported recently by Yao et al. [19] and Yamamoto et al. [20]. While both produced blend nanofiber of chitosan/poly(ethylenoxide) or poly(vinyl alcohol) from the chitosan solution mixed with PEO or PVA, the latter also reported that electrospinning of homogeneous chitosan nanofiber using trifluoroacetic acid/dichloromethane solvent, which requires additional extraction of the organic solvents. And it was also mentioned by the latter that chitosan solution dissolved in 0.2 M acetic acid and its solvent mixtures with various volatile organic solvents or aprotic solvents cannot produce chitosan nanofiber via electrospinning. However, we electrospun homogeneous nanofibers of pure chitosan dissolved in strongly concentrated aqueous acetic acid solution without addition of other solvents, which have not been reported yet. It is considered that the surface tension depression produced by increasing acetic acid concentration in water is the most important solution factors in the electrospinning of chitosan.

2. Experimental section

2.1. Materials and testing

Three kinds of demineralized and deproteinized chitosan powders were used as supplied from Taehun Bio Ltd., Korea. Reagent grade glacial acetic acid and deionized water were used for solvent. Viscometry was used for determining molecular weight of chitosan powder and nanofibers from intrinsic viscosity using a Mark–Houwink equation $[\eta] = KM_v^a$, where constants K and a of chitosan are $8.93 \times 10^{-2} \text{ cm}^3/\text{g}$ and 0.71, respectively, when 0.2 M acetic acid/0.1 M NaCl and 4 M urea were used as a solvent system [21]. A DV-II+ brookfield viscometer was used to measure the viscosity of chitosan/acid solution with No. 3 spindle at 25 °C after equilibrating for 5 min. A static surface tension meter was used to measure the surface tension of chitosan solution using Du Nouy ring method. Degree of deacetylation (DD) of chitosan was measured by a JASCO FT/IR-300E spectrophotometer according to the following equation: [22] $\text{DD} (\%) = (0.759 - 0.714 \times A_{1560}/A_{1070}) \times 100$. The DDs of three chitosan samples

with molecular weights of 30,000, 106,000 and 398,000 g/mol were 56%, 54%, and 65%, respectively. The morphology of the electrospun chitosan nanofibers were observed by a Hitachi S-4300 scanning electron microscope and the average diameter and diameter distribution of the electrospun chitosan mat were determined by an image analyzer.

2.2. Electrospinning

Chitosan solution was prepared using concentrated aqueous acetic acid solution with different amount of acetic acid in water at room temperature for 12 h and continuously supplied into a Hamilton gas-tight syringe with a feeding rate of 20 $\mu\text{L}/\text{min}$ using a syringe pump (KDS 220). The syringe used in the experiment had 20 gauge needle (capillary diameter, 0.58 mm) and a high-voltage DC generator was used to generate voltage up to 40 kV. The anode of the voltage power supply was connected with the tip of the syringe and the cathode attached to the plate covered by aluminum foil on which electrospun fibers were deposited. The electrospun samples were dried under vacuum at 60 °C overnight to remove acetic acid and water possibly remaining after the electrospinning.

3. Results and discussions

3.1. Acetic acid concentration effect

Generally, surface tension determines the upper and lower boundaries of electrospinning window if all other variables are held constant. The formation of droplets, bead and fibers can be driven by the surface tension of chitosan solution and lower surface tension of the spinning solution helps electrospinning to occur at lower electric field. Usually, chitosan solution can be easily made by dilute aqueous acid solution such as 1% or less and the chitosan solution containing dilute acetic acid was not electrospinnable irrespective of chitosan concentration. One of the most interesting results from our studies on the electrospun chitosan/acetic acid system is that the acetic acid concentration in water strongly influenced surface tension of chitosan solutions, which was remarkably important in chitosan electrospinning. As acetic acid concentration increased from 10% to 90%, surface tension decreased from 54.6 to 31.5 dyn/cm without significant viscosity change (Fig. 1). Net charge density of the chitosan solution also increased with increasing acetic acid concentration in water resulting in more charged ions available for charge repulsion. It has been known that salt addition in polymer solution increases charge density on the surface of ejected jet in electrospinning field and the higher charge density carried by the jet formed a smooth

fiber more likely due to stronger whipping instability of the jet [4]. Among the acetic acid concentrations studied, 90% aqueous acetic acid was found to be most suitable for the formation of uniform nanofiber as shown in Fig. 2. At the acetic acid concentration of 30% or more, several thin nanofibers about 40 nm in average diameter were initially produced together with large beads, and at 90% acid concentration the fibers diameter increased to 130 nm without beads. However, neat acetic acid did not dissolve enough chitosan to be electrospinnable. Therefore, the well-known difficulty in the electrospinning of chitosan nanofiber was solved by dissolving chitosan in strongly concentrated acetic acid in water resulting in lowering surface tension and increased charge density of aqueous chitosan solution. Therefore, the improved electrospinning behavior of the chitosan solution can be attributed to finding of appropriate acetic acid concentration. The decreased surface tension and increased

charge density of the jet at the same time resulted in lower electric field required for jet formation.

3.2. Chitosan concentration and molecular weight effect

In the electrospinning of certain molecular weight polymer, fibrous structure can be stabilized only above a critical concentration and below process limiting viscosity [23,24]. Molecular weight also has a significant effect on the rheological and electrical properties such as viscosity, surface tension, conductivity and dielectric strength. It has been reported previously that too low molecular weight polymer tend to form beads rather than fibers and high molecular weight polymers give fibers with larger average diameter [25,26]. Chitosan concentration of three different molecular weights closely related to viscosity and spinnability as shown in Fig. 3. Among three chitosans with different

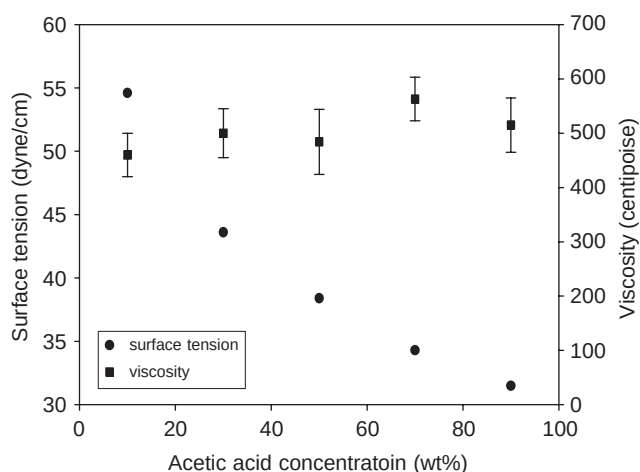


Fig. 1. Surface tension and viscosity of 7 wt% chitosan (106,000 g/mol) solution with different acetic acid concentrations.

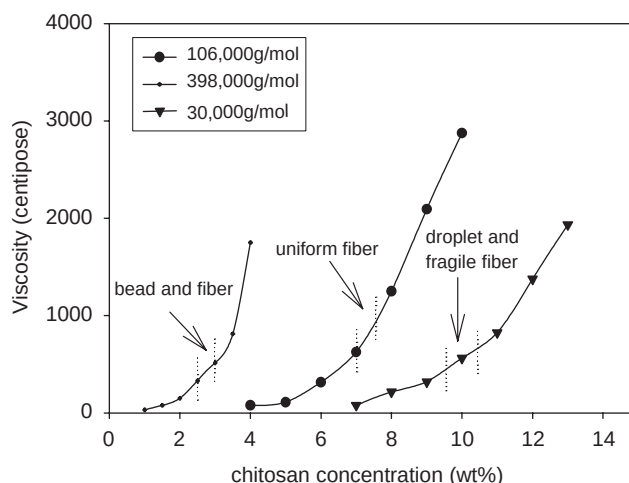


Fig. 3. Spinnability and relationship between concentration and viscosity of chitosans. Best spinnable conditions were between the concentrations indicated by vertical lines.

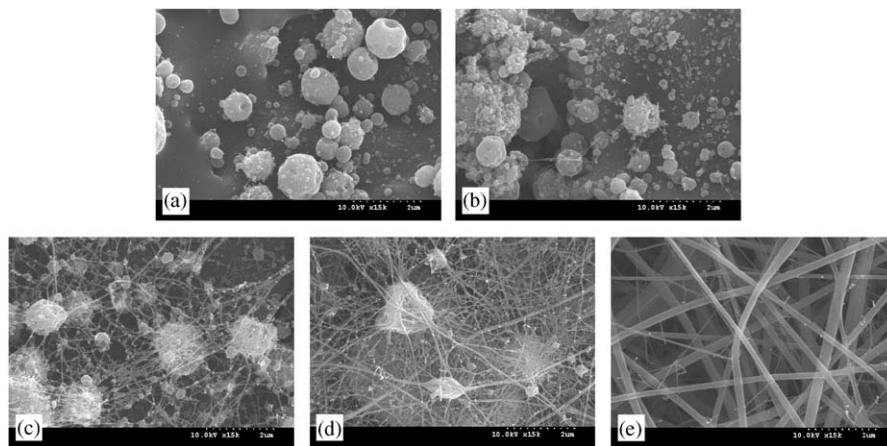


Fig. 2. Scanning electron micrographs of electrospun fibers at different acid concentration (wt%): (a) 10, (b) 30, (c) 50, (d) 70, (e) 90; 4 kV/cm, 7 wt% chitosan (106,000 g/mol).

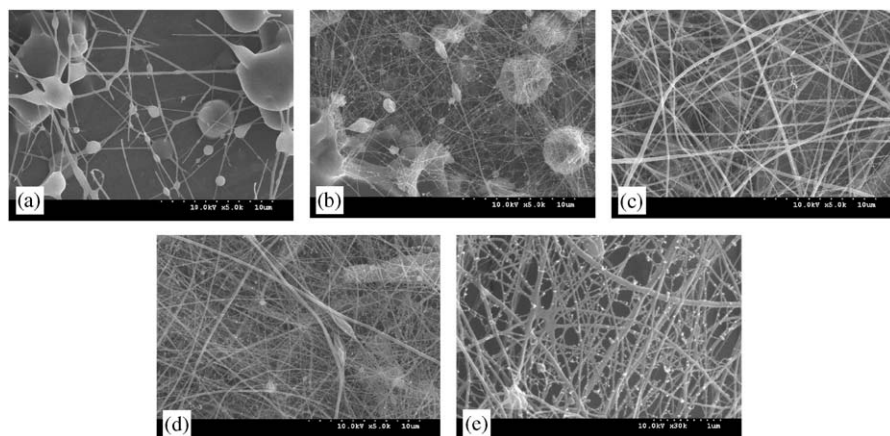


Fig. 4. Scanning electron micrographs of electrospun fiber at different molecular weights of chitosan: (a) 30,000 g/mol, 10%; (b) 106,000 g/mol, 6%; (c) 106,000 g/mol, 7%; (d) 106,000 g/mol, 8%; (e) 398,000 g/mol, 2.5%; 4 kV/cm, 90 wt% acetic acid, magnification: 5000 $\times$ (a, b, c, d); 30,000 $\times$ (e).

molecular weights and polymer concentrations, only about 7–7.5% of 106,000 g/mol chitosan (viscosity ranging from 484 to 590 cP) can produce a continuous and uniform fiber. The electrospun samples of the lower molecular weight chitosan solution (9.5–10.5%) usually contained large size beads and the fibers were fragile, while those of the higher molecular weight chitosan solution (2.5–3%) showed rougher and finer nanofibers with some bead defects. Fig. 4(a, c, e) shows the fiber morphology of three kinds of chitosan solution with similar viscosity at the most suitable spinning condition studied. With 30,000 g/mol chitosan only fragile fibers with several droplets were obtained. Uniform nanofiber morphology was stabilized at molecular weight of 106,000 g/mol, while very fine fibers (average diameter, 60 nm) with rougher surfaces was produced at 398,000 g/mol. The large population of smaller fiber at the highest molecular weight chitosan may be due to higher charge density of the chitosan molecules having higher amino groups per molecule. Drastic morphological changes were observed when chitosan concentration increased at the same molecular weight. With an increase in chitosan concentration, morphology seemed to change gradually from a mixture of droplet and fiber, through uniform fiber to fibers with spindle-like beads. At 6% concentration (Fig. 4(b)) chain entanglements may not be enough, resulting in spraying of large beads connected with very thin fibers. Fine fibers with some spindle-like beads were formed at 8% concentration (Fig. 4(d)) in comparison with bead-free fibers at 7% sample (Fig. 4(c)). Therefore 7% chitosan concentration, at a molecular weight of 106,000 g/mol, was the most suitable for uniform and bead-free fiber spinning.

3.3. Electric field effect

Once an electric field is applied, the surface of the polymer solution becomes charged. The transformation

of the drop at the needle tip to the fiber on the collector may be explained by spontaneous self-repulsion of the surface charges carried by the jet under electric field and the bending force brought by electric field. As most biomacromolecular nanofibers have been formed at the electric field range of 0.5–4 kV/cm [8–17], electric field more than 3 kV/cm was required for uniform chitosan nanofiber formation. Fig. 5 shows the electric field effect on the fiber morphology at the same spinning conditions.

At a electric field of 1 kV/cm spindle-like bead and thick fiber structure was formed (Fig. 5a). When the electric field was high enough to overcome the surface tension of chitosan solution, particularly in the range of 3–4.5 kV/cm or more, uniform and homogeneous fibers were formed. However, above 4.5 kV/cm much thinner fiber was produced with many bead defects possibly due to increased elongation force and instability of charged jet induced by the stronger electric field. In addition, different tip-to-collector distance and applied voltage at the same electric field did not affect the above-mentioned electric field effect significantly. Consistent nanofiber patterns at the same electric field were observed irrespective of different applied voltages and gaps as can be seen in Fig. 5(b–e, g, h). The insensitivity of the two parameters at same electric field to the spun fiber morphology indicates that the parameter was not an individual parameter for chitosan electrospinning as long as the enough distances was maintained for facile evaporation of the aqueous acetic acid solvent. Average fiber diameters of nanofiber samples electrospun under 3 and 4 kV/cm were 180 and 130 nm, respectively, and the diameter distribution became narrower with increasing electric field. As electric field increases from 1 to 5 kV/cm, average diameters decreased from 290 to 100 nm with increasingly narrower size distribution if the bead defects were not included for the calculation.

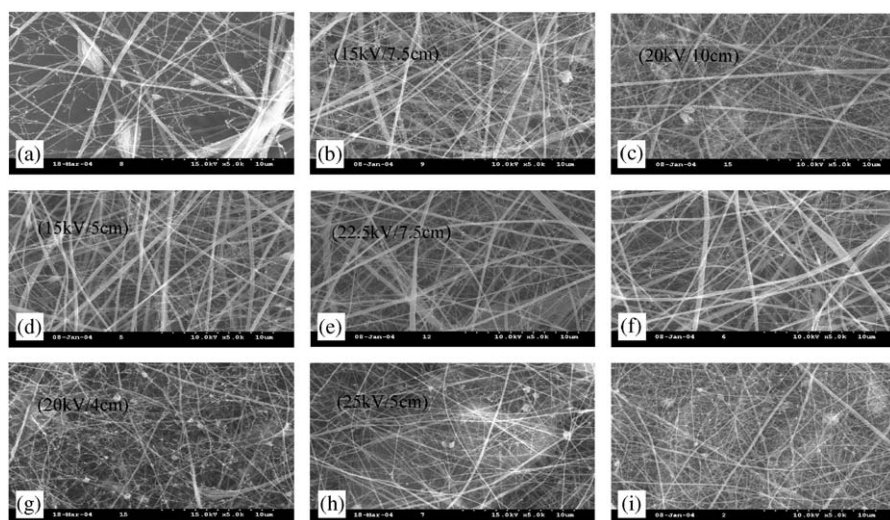


Fig. 5. Scanning electron micrographs of electrospun fibers at different electric fields (kV/cm): (a) 1, (b, c) 2, (d, e) 3, (f) 4.5, (g, h) 5, (i) 9; 7 wt% chitosan (106,000 g/mol), 90 wt% acetic acid.

4. Conclusions

Several parameters including molecular weight of chitosan, chitosan concentration, acid concentration and electric field were investigated in the electrospinning of chitosan solution dissolved in concentrated acetic acid solution. Among the parameters the acetic acid concentration was the most important parameter in the electrospinning, which decreased surface tension of the chitosan solution and at the same time increased charge density of jet without significant effect on viscosity. The fiber morphology changed from beaded fiber to uniform fiber with increasing acetic acid concentration, implying significant influence of the concentrated acetic acid on reducing the applied field required for electrospinning. Only chitosan solution of a molecular weight of 106,000 g/mol at 7% concentration, dissolved in 90% aqueous acetic acid solution, formed a bead-free and more uniform nanofiber. Electric field higher than 3 kV/cm but lower than 5 kV/cm was required to produce bead-free chitosan nanofiber and the average diameter decreased with narrower diameter distribution as applied electric field increased.

Acknowledgments

This research was supported by the Program for the Training of Graduate Students in Regional Innovation (KB-007, 2004), which was conducted by the Ministry of Commerce, Industry and Energy of the Korean Government.

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